Assembly and Characterization of a Galileoscope: Magnification, Field of View and Exit Pupil

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I. Introduction

The instrument characterized in this experiment is a Galileoscope, a kit refractor with a 50 mm achromatic objective of 500 mm focal length ($f/10$). The kit provides a 20 mm eyepiece, which gives a nominal magnification of $25\times$, and a $2\times$ Barlow tube that extends the effective focal length to give $50\times$ [1]. The telescope was assembled from the parts in the box following the supplied instructions, and then measured as an optical system: magnification M, field of view (FOV) and exit pupil diameter D_{ex}.

Two independent methods were used. The first is afocal photography: a 30 cm ruler was fixed to a tripod at a measured distance $L = 18.7\,\mathrm{m}$ from the telescope, and a phone camera photographed it twice, once directly and once through the eyepiece with the camera lens placed at the exit pupil. Because the camera settings are the same in both frames, the ratio of the pixel lengths of the same physical object gives the angular magnification directly,

$$M = \frac{\theta_i}{\theta_o} = \frac{d_{px,tel}}{d_{px,dir}}, \tag{1}$$

without any pixel-to-metre calibration. The same pair of frames also gives the field of view: the ruler spans the full illuminated field of the eyepiece, so the angular width of the field is

$$\mathrm{FOV} = 2\arctan\!\left(\frac{W}{2L}\right), \tag{2}$$

with $W = 30\,\mathrm{cm}$.

The second method uses the exit pupil. In an afocal system the magnification is the ratio of the entrance to the exit pupil diameters,

$$M = \frac{D_{obj}}{D_{ex}}. \tag{3}$$

The $25\times$ and $50\times$ configurations are Keplerian, so the exit pupil is real and can be projected: illuminating the objective and moving a sheet of paper behind the eyepiece produces a bright disc whose diameter is minimum at the exit pupil plane. This method is geometric and does not depend on the camera, so it acts as a cross-check of (1).

Section II lists the equipment, Section III reports the measurements, Section IV discusses the scatter and the systematic effects, and Section V summarizes the results.

II. Equipment, Materials and Instruments

Table I lists everything used during the session. The telescope and the target were mounted on separate tripods so that both could be aligned and left fixed for the whole set of repetitions.

TABLE I: Equipment used in the experiment.

Item	Use
Galileoscope (50 mm $f/10$)	instrument under test
Photographic tripod	telescope mount
Second tripod	target (ruler) mount
30 cm ruler (mm graduation)	target of known length
Measuring tape (30 m, mm)	telescope–target distance L
Vernier caliper (0.05 mm)	D_{obj} and D_{ex}
Phone camera	afocal and direct imaging
Phone flashlight	illumination of the objective
Sheet of white paper	screen for the exit pupil

III. Measurements

A. Magnification by afocal photography

The telescope was set at $L = 18.7\,\mathrm{m}$ from the ruler and five independent pairs of frames were taken with the $25\times$ eyepiece, re-centring the camera on the eyepiece between pairs. Fig. 1 shows one such pair. The length of the ruler was measured in pixels in both frames; the results are in Table II, where the per-trial uncertainty comes from propagating a ± 1 px reading error through (1).

TABLE II: Pixel length of the 30 cm ruler in the direct and afocal frames, and the resulting magnification.

Trial	$d_{px,dir}$ [px]	$d_{px,tel}$ [px]	M_i	σ_{M_i}
1	33	815	24.70	0.75
2	32	796	24.88	0.78
3	34	779	22.91	0.68
4	32	811	25.34	0.79
5	33	803	24.33	0.74
Mean $\bar{M}$				24.43
Sample deviation s				0.92
Error of the mean $s/\sqrt{N}$				0.41

The five values are plotted in Fig. 2. The result is

$$M_{25\times} = 24.43 \pm 0.41, \tag{4}$$

which sits 1.4σ below the nominal $25\times$.



(a) (b)

Fig. 1: (a) Direct frame: telescope on its tripod, the measuring tape along the ground and the target tripod at 18.7 m. (b) Afocal frame of the same target through the 25× eyepiece; the image is inverted and the ruler fills the illuminated field.

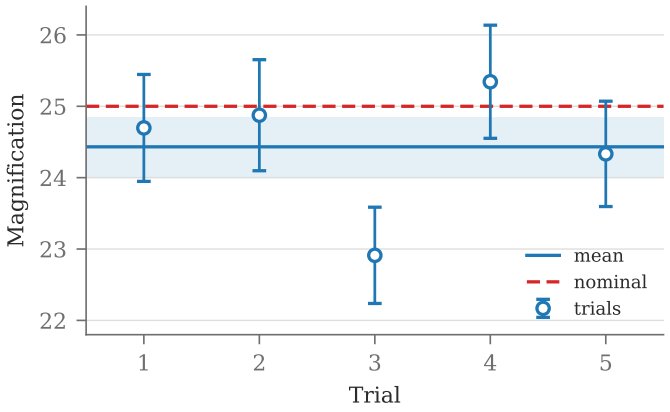


Fig. 2: Per-trial magnification with the ±1 px propagated error. The solid line is the mean and the band its standard error; the dashed line is the nominal value.

B. Field of view

In the afocal frames the 30 cm ruler spans the illuminated field of the eyepiece, so $W = 300 \pm 10$ mm is taken as the field width at $L = 18.7 \pm 0.3$ m. From (2),

$$\mathrm{FOV}_{25\times} = 16.04 \pm 0.59 \text{ mrad} = 0.919^\circ \pm 0.034^\circ, \quad (5)$$

that is $3309'' \pm 122''$. Combined with the measured magnification, the apparent field seen by the observer is $\mathrm{AFOV} = M \cdot \mathrm{FOV} = 22.5^\circ \pm 0.9^\circ$, a value consistent with a simple two-element eyepiece.

C. Exit pupil projection

The objective clear aperture was measured with the caliper, and the exit pupil was projected on a sheet of paper using the phone flashlight as the source, sliding the paper until the disc reached its minimum diameter. Five readings were taken for each quantity; only the 25× and 50× configurations were measured this way. The data are in Table III and compared with the expected values in Fig. 3.

TABLE III: Caliper readings of the objective aperture and of the projected exit pupil, in mm.

Quantity	1	2	3	4	5	Mean	$s/\sqrt{N}$
D_{obj}	47.1	47.0	47.2	47.3	47.1	47.14	0.05
D_{ex} (25×)	1.00	0.90	0.80	1.00	1.00	0.94	0.04
D_{ex} (50×)	0.50	0.40	0.60	0.66	0.52	0.54	0.04

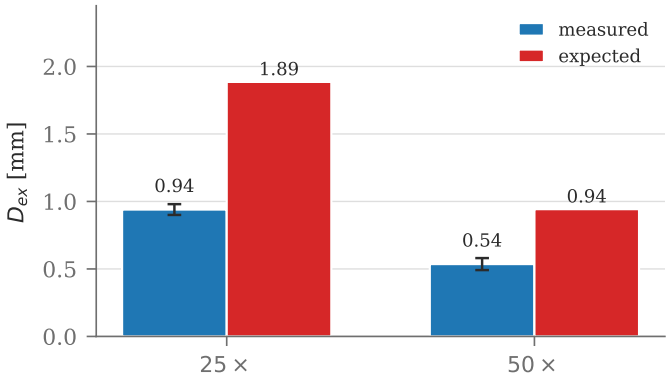


Fig. 3: Measured exit pupil against the value expected from the nominal magnification, D_{obj}/M_{nom}, for both Keplerian configurations.

Applying (3) with the relative errors added in quadrature,

$$M^{proj}_{25\times} = 50.2 \pm 2.1, \quad (6)$$

$$M^{proj}_{50\times} = 88.0 \pm 7.3. \quad (7)$$

Table IV collects the results of both methods.

TABLE IV: Summary of the measured parameters.

Parameter	**25×**	**50×**	**Nom.**
M (photographic)	24.43 ± 0.41	—	25 / 50
M (projection)	50.2 ± 2.1	88.0 ± 7.3	25 / 50
FOV	$0.919^\circ \pm 0.034^\circ$	—	—
D_{ex} [mm]	0.94 ± 0.04	0.54 ± 0.04	—
D_{obj} [mm]	47.14 ± 0.05		50

IV. Discussion

The photographic magnification agrees with the nominal value. Its scatter, $s/\bar{M} = 3.8\,\%$, is fully explained by the reading error of the direct frame: the ruler is only 32–34 px long there, so a ±1 px uncertainty is already 3 %, while the same error on the ∼ 800 px afocal measurement is negligible. Trial 3 carries most of the dispersion because it combines the largest direct reading with the smallest afocal one. The measurement is therefore limited by the angular sampling of the camera in the direct frame, not by the telescope; photographing a longer target, or the same target from closer, would reduce the scatter without changing the method.

The projection method behaves differently. It gives magnifications about twice the nominal ones, which is far outside the quoted errors. Two arguments show this is a bias of

the method as applied and not a property of the telescope. First, the photographic result for the same $25\times$ configuration predicts $D_{ex} = D_{obj}/\bar{M} = 1.93\,\mathrm{mm}$, while the projected disc measured 0.94 mm, a factor 2.05 smaller. Second, the ratio between the two configurations, $M^{proj}_{50\times}/M^{proj}_{25\times} = 1.75$, is close to the $2\times$ of the Barlow tube, so the relative behaviour is right and only the absolute scale is displaced.

The most likely cause is the illumination. The exit pupil is the image of the objective aperture formed by the eyepiece, and reproducing it requires the whole aperture to be filled with light. A phone flashlight is a small, bright and nearly point-like source placed close to the objective, so instead of filling the aperture it illuminates only a part of it, and the disc formed behind the eyepiece is closer to an image of the source than to the exit pupil. A diffuse extended source, such as the daytime sky or a white wall seen from a distance, would be the correct illumination. Two secondary effects push in the same direction: the disc has a soft edge, so the observer tends to measure the bright core rather than the full footprint, and the axial position of minimum diameter has to be found by eye, which biases the reading towards the waist of the beam.

The measured aperture, 47.14 mm, is 2.9 mm smaller than the 50 mm quoted for the lens. This is expected: the caliper reads the free aperture left by the plastic cell, not the diameter of the glass. Using this value instead of 50 mm lowers the projected magnifications by 6 % and is the more honest choice, since it is the aperture that actually collects light.

Several systematic effects remain in the photographic branch. The camera axis was aligned with the eyepiece by hand, so a residual tilt produces asymmetric vignetting and shortens the measured field. The distance L was read on a tape lying on the ground, and its origin is the tripod column rather than the entrance pupil of the objective; this is a fixed offset of a few centimetres, below the 0.3 m assigned to L. The ruler was assumed perpendicular to the optical axis, and any tilt foreshortens W and underestimates the FOV. Finally, the FOV value rests on the assumption that the 30 cm ruler fills the illuminated field; any mismatch transfers linearly to the FOV. This is the reason for the 10 mm assigned to W, which together with the 0.3 m on L gives the 3.7 % relative error quoted above and dominates that result.

The afocal measurements were taken only for the $25\times$ eyepiece, so the photographic magnification and the FOV of the $50\times$ configuration are not available, and the cross-check between methods could be done for one configuration only.

V. Conclusions

The Galileoscope was assembled and its $25\times$ configuration characterized by afocal photography, giving $M = 24.43 \pm 0.41$ and $\mathrm{FOV} = 0.919^\circ \pm 0.034^\circ$ ($3309'' \pm 122''$), from which an apparent field of $22.5^\circ \pm 0.9^\circ$ follows. The magnification agrees with the nominal value within 1.4σ, and its scatter is set by the ± 1 px reading of a 33 px target in the direct frame.

The exit pupil projection gave $D_{ex} = 0.94 \pm 0.04$ mm and 0.54 ± 0.04 mm for the $25\times$ and $50\times$ configurations, which translate into magnifications about twice the nominal ones. The internal ratio between configurations is correct (1.75 against 2), so the discrepancy is an absolute scale bias introduced by using a point-like flashlight, which underfills the objective, instead of an extended diffuse source. Repeating the projection against a uniformly illuminated white screen is the direct way to test this.

Between the two methods, the photographic one proved both more accurate and easier to control, since it depends on a ratio of pixel counts and not on locating a soft-edged disc by eye. The exit pupil measurement is still useful, but only as a relative comparison between configurations unless the illumination is corrected.

References

[1] Galileoscope, LLC, *Galileoscope Assembly Instructions*, IAU, 2009. www.galileoscope.org

[2] L. Vanzi, *Astronomical Instrumentation Lab IEE3873 – Galileoscope*, course guidelines, Pontificia Universidad Católica de Chile.